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Research Topic: Implementing Layer-Asymmetric Synaptic Pruning Derived from Hierarchical Neurodevelopment for Robust Deep Neural Networks

Title: Implementing Layer-Asymmetric Synaptic Pruning Derived from Hierarchical Neurodevelopment for Robust Deep Neural Networks

Introduction:

Brain development requires eliminating redundant neural connections through a process known as synaptic pruning [1]. Inspired by this biological mechanism, researchers have extensively integrated synaptic pruning principles into artificial neural network (ANN) architectures: substantially boosting resource efficiency, accuracy, and convergence speed [2], [3], [4]. Han et al. (2021) apply synaptic pruning through a framework called Developmental Plasticity-inspired Adaptive Pruning (DPAP), which mimics the "use it or lose it" principle of the developing human brain to optimize both Spiking Neural Networks (SNNs) and ANNs. The pruning process helped the network reach target accuracy significantly faster, with convergence speed improvements of up to 1.85x on specific datasets like N-MNIST [3].

In biological neural systems, however, pruning does not occur uniformly; primary sensory regions mature and undergo synaptic elimination far earlier than higher-order association areas [5]. Current pruning evaluations incorporate prevailing deep learning pruning paradigms such as global magnitude pruning, unstructured weight thresholding, or uniform layer-wise pruning which treat all network layers symmetrically or rely purely on static numerical magnitudes [6]. Consequentially, heavily compressed models often become structurally fragile, sacrificing representation depth and performing poorly when exposed to conditions outside of pristine training environments [7], [8]. As deep neural networks are progressively utilized in real-world, safety-critical environments; evaluating compression solely on clean, top-1 accuracy is insufficient [9].

This study addresses this gap by testing whether biologically aligned pruning yields superior generalization robustness compared to standard magnitude pruning. Generalization robustness will be systematically evaluated across three critical metrics: out-of-distribution (OOD) performance, adversarial perturbation resilience, and corrupted input tolerance. These metrics are used to test whether layer-asymmetric pruning preserves functional representation depth rather than overfitting to clean, i.i.d. training data; thereby extending model evaluation beyond standard top-1 accuracy.

Literature Review:

Although uniform parameter reductions had achieved massive compression rates, they often blindly truncate critical structural connections [13]. These limitations are recognized, but current non-uniform pruning approaches still remain mathematically constrained. Methods such as second-order Hessian-based techniques like Optimal Brain Surgeon [14] or iterative magnitude searches derived from the Lottery Ticket Hypothesis [15] rely heavily on static computational heuristics. These frameworks optimize strictly for parameter importance at a given convergence state rather than following a developmental schedule. Thus, they fail to model the temporal hierarchy of layer maturation. To introduce principled temporal dynamics, researchers have

progressively looked to neurobiology, observing that eliminating redundant synapses drives efficient learning and network regularization [1], [2]. Frameworks started to apply the biological "use it or lose it" mechanism. For example, Developmental Plasticity- inspired Adaptive Pruning (DPAP) dynamically optimize network structures, achieving sparsity levels exceeding 80% without degrading standard baseline accuracy [3], [4], [11].

In safety-critical applications, neural networks inevitably encounter unpredictable environmental noise and distributional shifts, making generalization robustness (the ability to maintain stable performance under non-ideal conditions) a paramount priority [9]. However, existing bio-inspired frameworks focus almost exclusively on pristine training data, and they largely overlook the distinct spatial and temporal hierarchy of genuine cortical development. In biological neural systems, primary sensory regions responsible for rapid visual feature integration mature quickly and undergo early synaptic pruning [10]. Conversely, higher-order association areas, such as the human frontal cortex, undergo prolonged periods of synaptic density expansion and demonstrate late-stage pruning [5], [12]. By mapping this biological hierarchy onto deep learning architectures (where shallow layers act as early sensory extractors and deep layers function as higher-order association spaces) it logically follows that shallow layers should be pruned earlier in the training cycle. Yet, existing literature has yet to systematically investigate how this specific temporal layer-asymmetry impacts structural model stability and generalization robustness.

Evaluating compressed networks purely on clean top-1 accuracy masks severe structural vulnerabilities, an issue increasingly prioritized in AI safety research [9]. Empirical studies demonstrate that standard global and magnitude-based pruning paradigms create brittle decision boundaries. While heavily pruned networks often maintain their baseline accuracy on pristine training data, their error rates on corrupted inputs, environmental noise, or underrepresented classes can disproportionately spike [7]. Moreover, aggressively pruning deeper layers prematurely reduces the model's capacity for complex, domain-invariant representations. This structural degradation not only leads to high out-of-distribution (OOD) error rates, but also leaves the network significantly more vulnerable to adversarial gradient attacks compared to dense baselines [8].

Currently, the existing literature lacks a comprehensive evaluation bridging hierarchical neurodevelopmental asymmetry with generalization robustness. This study fills this critical gap by systematically testing whether a biologically aligned, layer-asymmetric pruning schedule organically mitigates the structural fragility of compressed neural networks.

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Research Questions:

This study intends to answer the following primary research question: To what extent does integrating layer-asymmetric synaptic pruning dynamics (derived from hierarchical cortical neurodevelopment) enhance the generalization robustness of deep neural networks compared to standard uniform magnitude pruning? To aid in answering the preceding research question, three sub-questions are pursued,

I. *How does accelerating the pruning of low-level extraction layers affect the model's corrupted input tolerance to natural input corruptions?*
II. *Does maintaining structural density in deeper layers preserve abstract feature representations?*
III. *How does the adversarial perturbation resilience of a layer-asymmetrically pruned network compare to an identically compressed network pruned via standard global magnitude pruning?*

Proposed Methodology:

This method will take a quantitative approach.

Experimental Variables
The independent variable will be measured as the neural network pruning strategy applied: standard uniform global magnitude pruning in comparison to layer-asymmetric synaptic pruning. Generalization robustness will be utilized as the dependent variable, quantified across three specific metrics:

I. Mean Corruption Error (mCE) for natural input corruptions.
II. Out-of-Distribution (OOD) classification accuracy.
III. Adversarial accuracy under white-box gradient attacks.

Network Architecture Selection
Standard hierarchical convolutional networks, specifically ResNet-18 [16] and VGG-16 [17] will be employed.

These architectures possess traceable hierarchies suited for mapping neurobiology. The shallow convolutional blocks serve as the proxy for biological sensory extraction regions while the fully connected layers act as the higher-order association areas.

Evaluation Benchmarks
The following standardized benchmarks will be used to evaluate the models and establish reproducible baselines,

- CIFAR-10 and CIFAR-100 [18] to establish baseline top-1 accuracy on identically distributed (i.i.d.) data.

I. CIFAR-10-C and CIFAR-100-C [9] to calculate the mean Corruption Error (mCE). The benchmark applies 15 diverse types of natural environmental noise across 5 severity levels.
II. CIFAR-10.1 [19] to evaluate how well the preserved deep layers maintain domaininvariant feature representations compared to the uniform baseline by providing a distinct test set featuring a verified natural distributional shift.
III. Projected Gradient Descent (PGD) [20] to rigorously evaluate model resilience. This will be applied at varying perturbation magnitudes.

Pruning Algorithms

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As a control, an unstructured magnitude pruning baseline will be used. The algorithm removes the lowest-magnitude weights uniformly across all network layers throughout the training cycle until the target global sparsity is achieved [6], [13].

The layer-asymmetric pruning will proceed as followed,

- Phase 1 (Early Pruning): Aggressively applies magnitude-based pruning to the shallow extraction layers within the first ~30% of the training epochs, enforcing early stabilization of foundational filters.
- Phase 2 (Late Pruning): Delays the pruning of deeper representation layers until the final stages of training, applying a significantly lower sparsity threshold to preserve parameter density.

Project Practicalities:

This study will approximately span 14 weeks. The first 2 weeks will be dedicated to configuring the environment and establishing required evaluation benchmarks. Following this setup, weeks 3-5 will focus on training hierarchical networks to establish dense baselines, executing concurrently alongside a control model that utilizes uniform unstructured magnitude pruning. Subsequently, weeks 6-8 will be allocated to integrating and tuning the aforementioned layer-asymmetric pruning algorithm. Once the pruned models are optimized, weeks 9-11 will be dedicated to rigorous generalization robustness bench marking. Finally, weeks 12-14 will be reserved for aggregating the experimental data, analyzing performance metrics to validate structural stability, and drafting the final research manuscript.

Expected Outcome:

This study expects that the layer-asymmetric networks will match the uniform baseline's accuracy while demonstrating superior generalization robustness. The model should yield a lower Mean Corruption Error on CIFAR-10-C, maintain higher accuracy on the CIFAR-10.1 out-of-distribution benchmark, and display stronger resilience on the PGD. These results would validate that preserving deep-layer capacity produces robust, domain-invariant decision boundaries.

Limitations:

Although this study provides insights into biologically inspired pruning, multiple limitations exist. The evaluation relies on small-scale vision benchmarks. Findings from these datasets may not fully scale to high-resolution environments. Additionally, the scope is strictly restricted to standard convolutional neural networks: results cannot be directly projected to self-attention architectures.

Conclusion:

This research brings neurobiology and deep learning together through a layer-asymmetric pruning schedule. Preserving structural density in deep association layers while aggressively pruning shallow regions, this study attends to vulnerabilities in uniform pruning. Thus, the methodology promises resilient neural networks: capable of withstanding real-world noise and adversarial threats.

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